

Local AI Innovation and Firm Dynamics in U.S. Commuting Zones

Abstract

We study how local AI innovation is related to firm dynamics across U.S. commuting zones in 2005–2019 using PatentsView, Business Dynamics Statistics, County Business Patterns, and population data. In 721 commuting zones (10,815 CZ-year observations), higher AI patent intensity is associated with lower establishment exit (-0.0013 , SE 0.0005) and higher net entry (0.0019 , SE 0.0006); the entry estimate is positive but imprecise. The sign pattern is similar under alternative transformations of patent intensity and after excluding the most AI-intensive places. Because staggered-adoption event studies exhibit pre-trends and an auxiliary Bartik specification has a weak first stage ($F=3.39$), we treat the estimates as descriptive rather than causal. The evidence suggests that places with more AI patenting also experienced slightly stronger business dynamism, primarily through lower exit.

1 Introduction

Artificial intelligence is a plausible general-purpose technology: it diffuses across sectors, improves rapidly, and can reorganize production in both incumbent and entrant firms [Bresnahan and Trajtenberg, 1995, Helpman and Trajtenberg, 1998, Brynjolfsson et al., 2018]. For local economies, this raises a concrete empirical question: when AI-related innovation rises in a place, do we observe more establishment entry, more exit, or both?

We study that question using a commuting-zone panel for the United States over 2005–2019. The analysis combines inventor-location data from PatentsView with establishment dynamics from the Business Dynamics Statistics and baseline industry structure from County Business Patterns. Fixed-effects regressions are the core empirical design, while event-study and shift-share exercises are used only to gauge whether the data support a stronger interpretation.

The paper makes three contributions. First, it develops a simple framework linking AI diffusion to local firm dynamics. AI can lower entry costs for some firms while also improving incumbent adaptation, so the signs of exit and net entry are empirical rather than pinned down by theory. Second, it constructs a commuting-zone panel that links local AI patenting to establishment entry, exit, and job flows in a common geography over time. Third, it documents a consistent empirical pattern: places with more AI patenting tend to experience lower establishment exit and slightly higher net establishment entry, while the evidence for gross entry and job flows is weaker.

The main analysis covers 2005–2019 and includes 10,815 commuting-zone-year observations across 721 commuting zones. The treatment is the inverse hyperbolic sine of AI patents per 100,000 population. Because the event-study estimates fail the pre-trend criterion and the auxiliary Bartik specification yields a weak first stage (3.39), the results are interpreted as descriptive evidence on within-place co-movement rather than causal effects.

These results speak to several literatures. We extend the literature on automation and labor markets [Acemoglu and Restrepo, 2020, Webb, 2020] by focusing on firm dynamics

rather than employment. We contribute to work on local shocks and place-based adjustment [Autor et al., 2013, Moretti, 2010] by documenting how AI innovation co-moves with business dynamism. We also add to the growing literature on AI’s economic effects [Felten et al., 2021, Babina et al., 2024] with evidence that is explicit about the limits of identification.

2 Model

This section sketches a simple model of AI adoption and firm dynamics linking local AI innovation to establishment entry, exit, and net business formation. The framework combines elements of Schumpeterian competition [Schumpeter, 1942, Aghion et al., 2005] with the general-purpose technology literature [Bresnahan and Trajtenberg, 1995, Helpman and Trajtenberg, 1998], while allowing both entrants and incumbents to benefit from local AI spillovers.

2.1 Environment

Consider a local economy (commuting zone) with a continuum of industries $j \in [0, 1]$. Time is discrete, indexed by t . Each industry contains a single incumbent firm and a pool of potential entrants.

Technology. Firms produce output using a constant-returns technology. An incumbent in industry j produces with productivity A_{jt}^I . Potential entrants draw productivity A_{jt}^E from a distribution $F(\cdot)$ upon paying a fixed entry cost κ .

AI as a GPT. AI is a general-purpose technology that:

1. lowers some entry costs: $\kappa(\theta_t) = \bar{\kappa} - \gamma_\kappa \theta_t$, where θ_t is local AI intensity;
2. raises expected entrant productivity: $\mathbb{E}[A^E|\theta_t] = \bar{A} + \gamma_A \theta_t$;

3. can also lower incumbent adaptation costs, allowing some existing firms to survive longer.

The key empirical implication is that higher local AI intensity should encourage experimentation and entry, while the effect on exit depends on whether entrant displacement or incumbent adaptation is stronger.

2.2 Firm Decisions

Entry. A potential entrant pays $\kappa(\theta_t)$ to enter, draws productivity A^E , and operates if $A^E > \underline{A}$ (a minimum viable threshold). The entry rate is:

$$e_t = \lambda \cdot \Pr(A^E > \underline{A}) \cdot \mathbf{1}\{\pi^E(\theta_t) > \kappa(\theta_t)\} \quad (1)$$

where λ is the arrival rate of potential entrants and $\pi^E(\theta_t)$ is expected profit conditional on entry.

Exit. An incumbent exits when its continuation value falls below zero. AI affects that continuation value in two opposing ways. More productive entrants raise competitive pressure, but local AI spillovers may also improve incumbent productivity or reduce adoption costs. Exit therefore depends on whether displacement or adaptation dominates.

Incumbent Response. Incumbents can invest in AI adoption at cost $c(\theta_t)$, which is decreasing in local AI spillovers. Adoption raises incumbent productivity by factor $(1 + \delta)$. The adoption decision trades off the cost $c(\theta_t)$ against the benefit of surviving competitive pressure.

2.3 Equilibrium

In equilibrium, entry and exit rates are jointly determined by:

$$e^*(\theta) = \lambda \cdot [1 - F(\underline{A} - \gamma_A \theta)] \cdot \mathbf{1}\{\pi^E(\theta) > \bar{\kappa} - \gamma_\kappa \theta\} \quad (2)$$

$$x^*(\theta) = \Pr(V^I(\theta) < 0), \quad (3)$$

where $V^I(\theta)$ is the incumbent continuation value after accounting for both competitive pressure and adaptation opportunities.

2.4 Comparative Statics

Proposition 1 (Entry Effect). *If higher local AI intensity lowers effective entry costs and weakly raises expected entrant productivity, then equilibrium entry weakly rises over the region in which the profitability condition binds:*

$$\frac{\partial e^*}{\partial \theta} \geq 0 \quad (4)$$

Entry can be locally flat when the profitability indicator does not change, but it does not fall under these maintained assumptions.

Proposition 2 (Ambiguous Exit Effect). *Higher local AI intensity can increase or decrease establishment exit:*

$$\frac{\partial x^*}{\partial \theta} \geq 0 \quad (5)$$

Exit rises when entrant displacement dominates, and falls when incumbent adaptation benefits are larger.

Proposition 3 (Ambiguous Net Entry). *The sign of net entry is ambiguous:*

$$\frac{\partial(e^* - x^*)}{\partial \theta} \geq 0 \quad (6)$$

Net entry is positive when entry effects and incumbent survival dominate, and negative when displacement dominates.

Proposition 4 (Job Reallocation). *AI can affect job creation and job destruction through both entry and incumbent adjustment:*

$$\frac{\partial JC}{\partial \theta} \geq 0, \quad \frac{\partial JD}{\partial \theta} \geq 0 \quad (7)$$

Job creation is more likely to rise when entrant expansion dominates any contraction among incumbents.

2.5 Mapping to Empirical Specification

The model motivates the following estimating equation:

$$Y_{it} = \beta \cdot \text{AI}_{it} + \mathbf{X}'_{it}\gamma + \mu_i + \delta_t + \varepsilon_{it} \quad (8)$$

where Y_{it} is a firm-dynamics outcome, AI_{it} is local AI innovation intensity (the inverse hyperbolic sine of AI patents per capita), μ_i are commuting-zone fixed effects, δ_t are year fixed effects, and $\mathbf{X}_{it}$ collects controls interacted with trends.

The model predicts a positive coefficient for entry. Exit, net entry, and job flows remain empirical questions because AI can either intensify competition or improve incumbent adaptation.

2.6 Interpretation and Diagnostics

The fixed-effects estimate of β may still be biased by reverse causality or omitted local shocks. A stronger design would require either a credible quasi-experiment or an instrument with a demonstrably strong first stage.

We therefore use the baseline panel to characterize within-commuting-zone correlations

and treat auxiliary designs only as diagnostic checks on whether stronger causal language is warranted. One such exercise is a Bartik shift-share construction:

$$Z_{it} = \sum_j s_{ij,2005} \cdot g_{-i,jt}, \quad (9)$$

where $s_{ij,2005}$ is CZ i 's employment share in industry j in 2005 and $g_{-i,jt}$ is leave-one-out national AI patent growth by industry.

In our data, the Bartik first stage is weak, so that exercise does not materially sharpen identification.

3 Data

3.1 Commuting-Zone Geography

The unit of analysis is the 1990 commuting zone. County-to-commuting-zone and PUMA-to-commuting-zone crosswalks are drawn from David Dorn's public geography files, with the main sample restricted to the lower 48 states. County-level public data are aggregated to commuting zones before rates are computed. Because the public BDS county API uses Connecticut planning-region identifiers that do not align cleanly with the fixed county-to-CZ crosswalk, the sample excludes the Connecticut commuting zone.

3.2 AI Patent Intensity

The treatment is constructed from observed AI-related patents in PatentsView. Patents are restricted to utility grants in 2005–2019 and flagged as AI when at least one CPC code begins with one of the pre-declared AI prefixes: G06N3, G06N5, G06N7, G06N20 (neural networks and machine learning), G06F40 (natural language processing), G06V (image recognition), G06T7 (image analysis), B25J9/16 (adaptive control in robotics), and G10L15/17 (speech recognition). Inventor locations are assigned fractionally across inventor counties,

then aggregated to commuting zones and normalized by population.

The final treatment variable is $\text{asinh}(\text{AI patents per 100,000 population})$. Using the inverse hyperbolic sine preserves zero-patent places while dampening the right tail of the patent distribution.

3.3 Firm Dynamics

Firm-dynamics outcomes come from the Census Bureau’s Business Dynamics Statistics. We use county-level counts of establishment births, establishment exits, job creation, job destruction, establishments, and employment. These counts are aggregated to commuting-zone-year cells before entry, exit, and job-flow rates are computed.

3.4 Baseline Controls and Industry Structure

Baseline controls come from 2005 County Business Patterns. We construct 2005 manufacturing share, 2005 establishment counts, 2005 employment, and pre-period industry employment shares at the 2-digit NAICS level. These shares enter the baseline controls and the shift-share specification reported in the appendix.

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	P10	Median	P90
AI Patents per 100k	0.3206	1.6264	0.0000	0.0000	0.7642
$\text{asinh}(\text{AI Patents per 100k})$	0.1949	0.4666	0.0000	0.0000	0.7045
Entry Rate	0.0876	0.0222	0.0641	0.0849	0.1156
Exit Rate	0.0878	0.0204	0.0670	0.0859	0.1104
Net Entry Rate	-0.0001	0.0235	-0.0245	0.0000	0.0235
Job Creation Rate	0.1230	0.0302	0.0919	0.1192	0.1575
Job Destruction Rate	0.1210	0.0403	0.0868	0.1147	0.1598
Manufacturing Share (2005)	0.1429	0.0993	0.0146	0.1330	0.2693
High-Tech Share (2005)	0.2160	0.0753	0.1305	0.2077	0.3055
Log Population (2005)	11.5810	1.6420	9.3921	11.5915	13.6441
Observations	10,815				
Commuting Zones	721				
Years	2005–2019				

Notes: Unit of observation is a commuting zone $\times$ year. The treatment is the inverse hyperbolic sine of AI patents per 100,000 population. Firm-dynamics outcomes are aggregated from county-level BDS counts to the commuting-zone level before rates are computed.

4 Empirical Strategy

4.1 Baseline Fixed-Effects Specification

The core specification is a commuting-zone panel regression:

$$Y_{it} = \beta \text{AI}_{it} + \gamma_1 (\log \text{Pop}_{i,2005} \times t) + \gamma_2 (\text{MfgShare}_{i,2005} \times t) + \mu_i + \delta_t + \varepsilon_{it}, \quad (10)$$

where Y_{it} is a firm-dynamics outcome, AI_{it} is $\text{asinh}(\text{AI patents per } 100,000)$, μ_i are commuting-zone fixed effects, and δ_t are year fixed effects. Standard errors are clustered by commuting zone.

4.2 Interpretation and Diagnostics

Even with commuting-zone and year fixed effects, these estimates need not have a causal interpretation. Local conditions that foster AI patenting may also affect firm dynamics, and unobserved shocks may move both at the same time.

We therefore complement the baseline panel with two diagnostic exercises that speak to identification in this setting. The first is a staggered-adoption event-study specification. The second is a Bartik shift-share construction:

$$Z_{it} = \sum_j s_{ij,2005} \cdot g_{-i,jt} \quad (11)$$

where $s_{ij,2005}$ is CZ i 's 2005 employment share in industry j and $g_{-i,jt}$ is leave-one-out national AI patent growth by industry.

In our data, neither exercise materially strengthens identification. The event-study estimates exhibit pre-trends for the main outcomes, and the Bartik specification has a weak first stage (3.39). Accordingly, the main estimates are interpreted as descriptive within-CZ associations.

5 Results

5.1 Main Panel Estimates

Table 2 reports the baseline fixed-effects estimates. Higher AI patent intensity is associated with lower establishment exit (-0.0013, SE 0.0005) and higher net establishment entry (0.0019, SE 0.0006). The negative exit association is the strongest and most stable pattern in the table; the positive net-entry coefficient partly reflects that decline in exit. The entry coefficient is positive but imprecise (0.0007, SE 0.0004), while the job-flow estimates are smaller and less precisely estimated. Given the identification limits discussed above, these results are best read as within-CZ associations over time.

Table 2: AI Innovation and Firm Dynamics

	(1)	(2)	(3)	(4)	(5)
	Entry Rate	Exit Rate	Net Entry Rate	Job Creation Rate	Job Destruction Rate
asinh(AI Patents/100k)	0.00067 (0.00043)	-0.00128** (0.00053)	0.00195*** (0.00064)	0.00118* (0.00065)	-0.00093 (0.00111)
CZ Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Baseline Controls $\times$ Trend	Yes	Yes	Yes	Yes	Yes
R^2 (within)	0.001	0.001	0.001	0.010	0.006
Observations	10,815	10,815	10,815	10,815	10,815

Notes: Each column reports the coefficient on the inverse hyperbolic sine of AI patents per 100,000 population from a separate commuting-zone panel regression with CZ and year fixed effects. Controls are 2005 log population and 2005 manufacturing share, each interacted with a linear year trend. Standard errors are clustered by commuting zone. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2 Identification Diagnostics

Table 3 shows that the event-study evidence does not justify a stronger interpretation: none of the specifications satisfies the pre-trend criterion, and for every outcome at least one pre-treatment lead is statistically distinguishable from zero.

The Bartik exercise points in the same direction. Its first stage is 3.39, well below conventional rules of thumb for a strong instrument, so the appendix tables are best viewed as diagnostic results rather than as a basis for causal inference.

Table 3: Event-Study Pre-Trend Diagnostics

Outcome	Treated CZs	Pre-period Leads	All Lead $p > 0.05$	Meets Pre-Trend Criterion
Entry Rate	265	4	No	No
Exit Rate	265	4	No	No
Net Entry Rate	265	4	No	No
Job Creation Rate	265	4	No	No
Job Destruction Rate	265	4	No	No

Notes: Event studies use a staggered-adoption saturated specification. Treatment onset is defined as the first year, from 2012 onward, in which a CZ’s asinh AI patent intensity exceeds its 2005–2011 mean by one standard deviation for two consecutive years. We regard the diagnostic as satisfactory when at least three pre-treatment lead estimates are available and all reported lead coefficients have $p > 0.05$.

6 Heterogeneity

Table 4 asks whether the baseline pattern is concentrated in particular types of places. These split-sample results are descriptive and often imprecise, so they are best read as suggestive differences in where the correlations are most visible.

6.1 By Population Size

We split the sample at the median of log population in 2005. The net-entry association is more visible in smaller commuting zones, while the large-CZ coefficients are closer to zero and imprecise.

6.2 By Technology-Oriented Employment Share

We define the technology-oriented sector share using Information (NAICS 51), Professional Services (54), and Manufacturing (31–33), capturing both digital services and advanced production activities. The data do not show stronger net-entry associations in places with a larger initial technology-oriented employment base; if anything, the negative exit and positive net-entry estimates are more visible in the lower-share half of the sample.

6.3 By Initial AI Intensity

We split by 2005 AI patent intensity to distinguish places with an earlier AI footprint from those starting closer to zero. Entry is more positively associated with AI in higher-initial-AI CZs, but the net-entry association is clearer in lower-initial-AI CZs because the exit coefficient is more negative there.

Table 4: Heterogeneity by Commuting-Zone Characteristics

	Entry Rate	Exit Rate	Net Entry Rate
<i>By Population Size</i>			
Above-Median Population	0.00066 (0.00055)	0.00004 (0.00054)	0.00062 (0.00071)
Below-Median Population	0.00017 (0.00065)	-0.00177* (0.00090)	0.00194** (0.00097)
<i>By Technology-Oriented Employment Share</i>			
Above-Median Tech-Oriented Share	0.00048 (0.00053)	-0.00012 (0.00070)	0.00061 (0.00067)
Below-Median Tech-Oriented Share	0.00049 (0.00060)	-0.00198*** (0.00076)	0.00247** (0.00097)
<i>By Initial AI Intensity</i>			
Above-Median AI Intensity (2005)	0.00146** (0.00068)	0.00076 (0.00080)	0.00071 (0.00078)
Below-Median AI Intensity (2005)	0.00006 (0.00051)	-0.00123* (0.00064)	0.00128* (0.00078)

Notes: Each cell reports the coefficient on $\text{asinh}(\text{AI patents per 100k})$ from a separate regression estimated on the indicated subsample. All specifications include commuting-zone and year fixed effects plus the baseline controls interacted with a linear year trend. Standard errors are clustered by commuting zone. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7 Robustness

The core sign pattern is also robust to simple changes in specification and sample.

7.1 Alternative Treatment Variables

We re-estimate using $\log(1 + \text{AI patents per 100k})$ instead of the inverse hyperbolic sine transformation. The sign pattern is unchanged.

7.2 Excluding Outliers

We exclude commuting zones in the top 5% of mean AI patent intensity to ensure results are not driven by a handful of tech hubs. Net entry remains positive and exit remains negative.

Table 5: Robustness Checks

Specification	Entry Rate	Exit Rate	Net Entry Rate
Baseline	0.00067 (0.00043)	-0.00128** (0.00053)	0.00195*** (0.00064)
Log(1+AI)	0.00083 (0.00056)	-0.00161** (0.00067)	0.00244*** (0.00082)
Excl. Top 5	(0.00043)	(0.00053)	(0.00067)

Notes: Each row reports coefficients from an alternative specification. All regressions include commuting-zone and year fixed effects plus the baseline controls interacted with a linear year trend. “Log(1+AI)” uses $\log(1 + \text{AI patents per 100k})$ instead of asinh . “Excl. Top 5%” excludes CZs in the top 5% of mean AI patent intensity. Standard errors are clustered by commuting zone. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8 Discussion

8.1 Economic Magnitude

The estimated net-entry association is modest relative to gross entry and exit rates, both of which average about 0.088 and 0.088. The central pattern is therefore incremental rather than transformative: places with more AI patenting also exhibit slightly stronger net business dynamism, driven mainly by lower exit.

8.2 Interpretation

Because exit falls rather than rises in the baseline estimates, the results do not fit a strong creative-destruction narrative. A more cautious reading is that, during 2005–2019, places with more local AI patenting also saw somewhat better firm survival and slightly higher net establishment growth. The estimates do not isolate the mechanism behind that pattern, which could reflect AI-specific adaptation, complementary local investments, or other shocks correlated with patenting.

8.3 Policy Implications

Even so, the pattern may matter for regional inequality and local economic development. The present design, however, is better suited to documenting that co-movement than to evaluating policy interventions.

9 Data Availability

All data used in the main analysis are publicly available: David Dorn’s commuting-zone crosswalks, Census county population estimates, the Census Bureau’s Business Dynamics Statistics, County Business Patterns, and PatentsView granted patent files. An extension to worker-level labor-market outcomes would additionally require an IPUMS USA extract, which is not used in the results reported here.

10 Conclusion

This paper documents a modest positive relationship between local AI patenting and net establishment growth in U.S. commuting zones, driven mainly by lower exit. The pattern survives several basic robustness checks and is most visible in the exit margin rather than in gross entry or job flows.

At the same time, the evidence does not support a causal interpretation. The staggered-adoption event studies exhibit pre-trends, and the Bartik shift-share instrument has a weak first stage. The contribution is therefore to document how AI-related innovation and local business dynamism moved together over 2005–2019 and to clarify where stronger identification would be needed.

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A Additional Tables

A.1 ACS Coverage and Labor-Market Extension

Labor-market outcomes are not reported because the corresponding IPUMS USA microdata were not available. If those files are added, the coverage diagnostics below indicate the sample support for extending the analysis.

A.2 Bartik Diagnostics

The appendix reports the shift-share construction as an auxiliary diagnostic exercise. Because the first stage is weak (3.39), these estimates do not support a causal interpretation.

Table 6: Bartik IV Estimates

	(1)	(2)	(3)	(4)	
	Entry Rate	Exit Rate	Net Entry Rate	Job Creation Rate	Job Destruction Rate
<i>Panel A: 2SLS Estimates</i>					
asinh(AI Patents/100k)	-0.02481 (0.01723)	0.03177 (0.02244)	-0.05658 (0.03633)	-0.06476 (0.03951)	0.00000 (0.00000)
<i>Panel B: First Stage Diagnostics</i>					
First-stage F				3.39	
Exceeds Rule-of-Thumb Threshold ($F > 10$)				No	
CZ Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Baseline Controls $\times$ Trend	Yes	Yes	Yes	Yes	Yes
Observations	6,405	6,405	6,405	6,405	6,405

Notes: These 2SLS estimates are reported as a supplementary specification. asinh(AI patents per 100k) is instrumented with a Bartik shift-share measure built from 2005 industry shares and leave-one-out national AI patent growth. The first stage is weak ($F=3.39$), so the coefficients are not interpreted causally. Standard errors are clustered by commuting zone. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7: Bartik First-Stage Diagnostics

	asinh(AI Patents/100k)
Bartik AI Instrument	0.0463* (0.0252)
First-stage F-statistic	3.39
Rule-of-Thumb Threshold ($F > 10$)	Yes
Exceeds Threshold	No
Instrument mean	0.9765
Instrument std. dev.	1.3198
Observations	6,405

Notes: The Bartik instrument is constructed as $\sum_j s_{ij,2005} \times g_{-i,jt}$, where $s_{ij,2005}$ is CZ i 's employment share in industry j in 2005 and $g_{-i,jt}$ is leave-one-out national AI patent growth by industry. This table is reported as a supplementary diagnostic. Because the first-stage F-statistic is below the conventional $F > 10$ rule of thumb, the manuscript does not use this design for causal interpretation. Standard errors are clustered by commuting zone.

Table 8: Rotemberg Weights: High-Leverage Industries

NAICS	Total Share	Avg. Growth	Rotemberg Weight	High-Tech
31-33	103.1624	4.4487	0.7292	Yes
54	24.4941	4.7019	0.1830	Yes
51	11.7509	4.7019	0.0878	Yes

Notes: Rotemberg weights identify industries with high leverage in the exploratory Bartik instrument. Total share is the sum of 2005 employment shares across all CZs. Avg. growth is the mean leave-one-out national AI patent growth rate. Under the current CPC-to-NAICS concordance, only three industries receive positive weight and manufacturing dominates the design. High-tech industries are NAICS 51 (Information), 54 (Professional Services), and 31-33 (Manufacturing).